

Week 10 - Radiosondes: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

The table below is a synthetic radiosonde profile designed for practice. Wind direction is the direction *from* which the wind blows. Treat the measurements as exact unless the question asks about uncertainty.

Height (m)	Pressure (hPa)	T ($^{\circ}\text{C}$)	RH (%)	Wind dir. ($^{\circ}$)	Speed (m s^{-1})
0	1010	14	75	220	4
500	955	10.5	82	230	6
1000	900	7	91	240	8
1500	845	3.5	97	250	10
2000	795	0	93	260	12
3000	700	-6	65	270	16
5000	540	-19	40	280	22
8000	360	-40	30	290	30
10000	265	-54	25	300	38
12000	195	-53	20	310	42

YOU SHOULD TRY Question 1: Which quantities are measured directly and which are derived?

Using the columns in the sounding table:

1. Identify the quantities measured by sensors carried on the radiosonde.
2. Explain how wind speed and direction are obtained during the flight.
3. Height is often calculated from pressure/GPS information rather than measured by a mechanical “height sensor”. State one reason why it is useful to report the profile versus both height and pressure.

Worked answer

Modern radiosondes directly measure pressure, air temperature and humidity using onboard sensors. The sonde position is tracked, normally by GPS; successive horizontal positions divided by elapsed time give wind velocity and hence wind speed and direction.

Meteorology frequently uses pressure coordinates because pressure is closely related to the mass of atmosphere above a level and is the natural vertical coordinate of many weather analyses. Geometric height is useful for relating the sounding to terrain, aviation, cloud-base height and physical distance. Reporting both connects dynamical/thermodynamic interpretation to geometric altitude.

YOU SHOULD TRY Question 2: Diagnose the thermodynamic structure of the sounding

Use the table to answer all parts.

1. Calculate the mean environmental lapse rate from the surface to 2.0 km.
2. Compare it with the DALR, 9.8 K km^{-1} , and state whether an unsaturated parcel is dry-statically stable through this layer.
3. Identify the layer most likely to contain cloud from the RH observations.
4. Estimate the freezing level.
5. Identify the main dry layer immediately above the likely cloud.
6. What temperature feature between 10 and 12 km suggests the tropopause has been reached?

Worked answer

Temperature falls from 14°C at the surface to 0°C at 2 km, so

$$\Gamma_{env} = \frac{14}{2} = 7.0 \text{ K km}^{-1}.$$

Because $7.0 < 9.8$, an unsaturated lifted parcel cools faster than its environment and tends to become colder/denser, so the layer is stable with respect to dry convection.

The highest RH occurs from roughly 1 to 2 km, especially around 1.5 km where RH is 97%, so this is the most likely cloud layer. The profile reaches 0°C at about 2 km, giving a freezing level near 2.0 km. RH then drops to 65% at 3 km and 40% at 5 km, indicating a pronounced dry layer above the cloud.

Temperature decreases to -54°C at 10 km but then increases slightly to -53°C by 12 km. This strong reduction/reversal of lapse rate is characteristic of the tropopause/lower-stratosphere transition.

YOU SHOULD TRY Question 3: Quantify the vertical wind shear

Between the surface and 10 km:

1. By how much does wind speed increase?
2. What is the mean increase in wind speed per kilometre over this depth?
3. By how many degrees does the reported wind direction change, and does it veer clockwise or back anticlockwise with height?
4. State one forecasting reason why this vertical change in wind cannot be inferred from surface observations alone.

Worked answer

Wind speed increases from 4 to 38 m s^{-1} , a change of

$$\Delta V = 34 \text{ m s}^{-1}.$$

Over 10 km, the mean speed increase is

$$\frac{34}{10} = 3.4 \text{ m s}^{-1} \text{ km}^{-1}.$$

Wind direction changes from 220° to 300° , a clockwise change of 80° , so the wind veers with height. (A full vector-shear calculation would use the horizontal wind components rather than just speed and direction differences.)

Upper-air winds influence frontal structure, jet streams, storm motion and divergence/convergence aloft. A weak surface wind therefore does not imply weak flow through the rest of the troposphere.

EXTRA CHALLENGE Question 4: Why does the balloon expand and burst?

A weather balloon has volume 2.0 m^3 at launch, where $p_1 = 1010 \text{ hPa}$ and $T_1 = 287 \text{ K}$. Near the top of the profile, take $p_2 = 200 \text{ hPa}$ and $T_2 = 220 \text{ K}$. Neglect gas leakage and assume ideal-gas behaviour.

1. Calculate the balloon volume at the upper level using $pV/T = \text{constant}$.
2. By what factor has its volume increased?
3. If the envelope fails at about 8.0 m^3 , is a burst near this level plausible?

Worked answer

$$V_2 = V_1 \frac{p_1 T_2}{p_2 T_1} = 2.0 \frac{1010}{200} \frac{220}{287} = 7.74 \text{ m}^3.$$

The expansion factor is

$$\frac{7.74}{2.0} = 3.87.$$

The calculated volume is very close to the assumed 8.0 m^3 failure volume, so a burst near this pressure level is plausible. Real burst altitude also depends on balloon elasticity, gas temperature, ascent rate and manufacturing variability.

EXTRA CHALLENGE Question 5: Estimate ascent time and horizontal drift

Suppose the balloon ascends at a constant 5.0 m s^{-1} to 10 km . For a rough drift estimate, take the layer-mean horizontal wind speed to be 20 m s^{-1} .

1. How long does the ascent to 10 km take in seconds and minutes?
2. How far horizontally would the balloon travel at the assumed mean wind speed?
3. Give two reasons why the real trajectory would differ from this straight-line estimate.

Worked answer

The ascent time is

$$t = \frac{10000}{5.0} = 2000 \text{ s} = 33.3 \text{ min}.$$

The horizontal drift is approximately

$$d = Vt = 20(2000) = 40000 \text{ m} = 40 \text{ km}.$$

The real wind speed and direction vary strongly with height, as the sounding itself shows, and the balloon ascent rate is not perfectly constant. Turbulence and mesoscale flow variations also make the trajectory curved rather than a straight line.

YOU SHOULD TRY Question 6: Use the sounding for an operational forecast

A precipitation system is approaching the launch site. Use only the sounding information to assess the following statements as *supported*, *not supported*, or *cannot be decided from the sounding alone*. Give a brief reason for each.

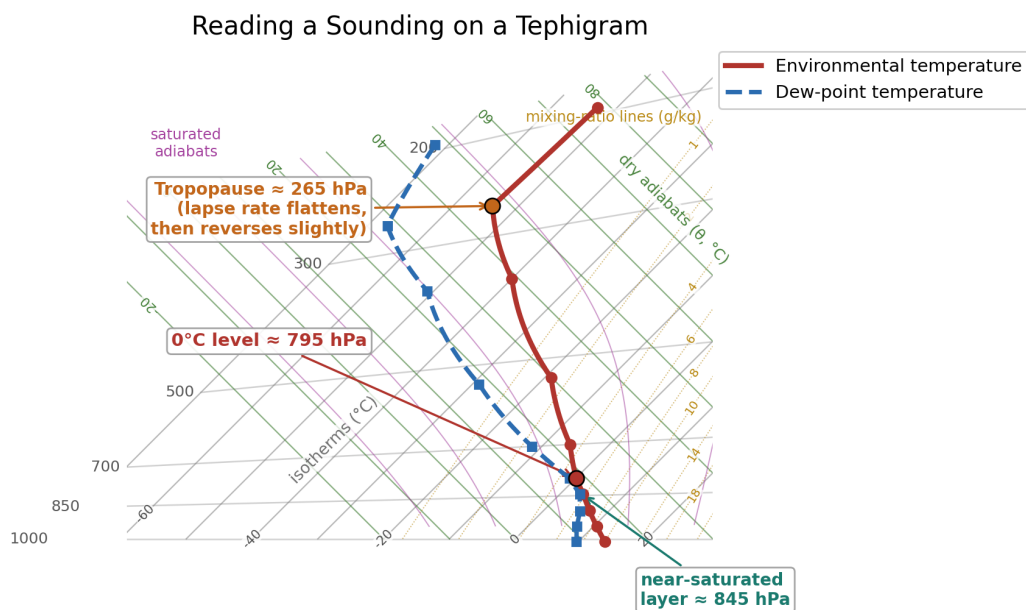
1. Low cloud is likely between about 1 and 2 km .
2. Any snow generated above 3 km must reach the surface as snow.
3. Aircraft could encounter a freezing level near 2 km .
4. Stronger winds occur aloft than at the surface.
5. The sounding proves that heavy rain will occur during the next hour.

Worked answer

1. Supported: RH is 91–97% through much of 1–2 km, strongly suggesting cloud or near-cloud conditions.
2. Not supported: temperatures are above freezing below 2 km, so falling snow/ice can melt before reaching the surface. Surface precipitation type depends on the full thermal profile and particle history.
3. Supported: $T = 0^\circ\text{C}$ at about 2 km.
4. Supported: wind speed rises from 4 m s^{-1} at the surface to more than 30 m s^{-1} aloft.
5. Cannot be decided: a sounding diagnoses the vertical atmospheric state at one time/place but does not by itself establish the intensity, timing or even occurrence of an approaching precipitation system. Radar, satellite, surface charts and forecasts are also needed.

IN CLASS Question 7: Read a plotted sounding rather than a table

The same synthetic sounding is plotted below on a simplified tephigram-style thermodynamic diagram. The solid purple curve is temperature and the dashed black curve is dew point. Grey guide lines show isotherms, dry adiabats and pressure levels. Use the diagram together with the table.



1. Identify the likely cloud layer from the separation between temperature and dew point.
2. Estimate the freezing level and tropopause height.
3. Where is the driest lower/mid-tropospheric air evident?
4. Explain one advantage of seeing the profile graphically rather than only as a table of numbers.

Worked answer

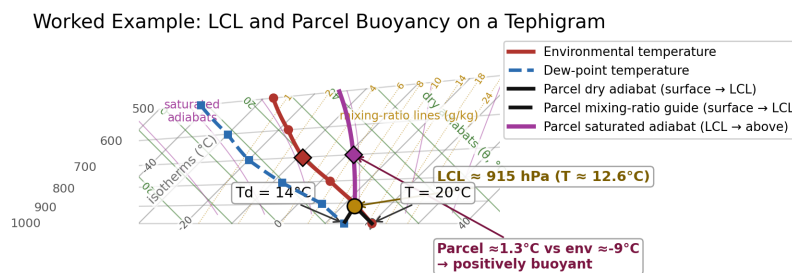
Temperature and dew point lie closest together from roughly 1 to 2 km, indicating near-saturation and likely cloud. The freezing level is close to 2 km where the temperature profile crosses 0°C . The temperature lapse rate becomes very small and reverses near 10–12 km, indicating the tropopause transition. The large separation between temperature and dew point above about 3 km indicates a pronounced dry layer.

A plotted sounding makes vertical structure immediately visible: cloud layers, inversions, freezing levels, dry layers and tropopause changes can be diagnosed by pattern rather than by repeatedly

comparing individual table entries. A thermodynamic diagram makes the same structure visible while also providing coordinates that can be used to compare environmental profiles with parcel paths and adiabatic changes.

Worked example: finding the LCL and testing parcel buoyancy on a tephigram. Q7 used a tephigram-style diagram to read off an environmental profile. A tephigram can also be used to trace what happens to a *rising parcel*, which is the skill behind lifting-condensation-level and stability questions more generally.

A surface parcel has $T = 20^\circ\text{C}$ and $T_d = 14^\circ\text{C}$ at 1000 hPa (here, the same as the environment at the surface). The figure below draws two guide lines from this starting point: the parcel's *dry adiabat* (solid black), along which an unsaturated parcel cools at the dry adiabatic lapse rate, and its *mixing-ratio guide* (dash-dot black), along which the parcel's dew point falls as pressure drops, because its water-vapour mixing ratio r_v stays fixed until the parcel saturates.



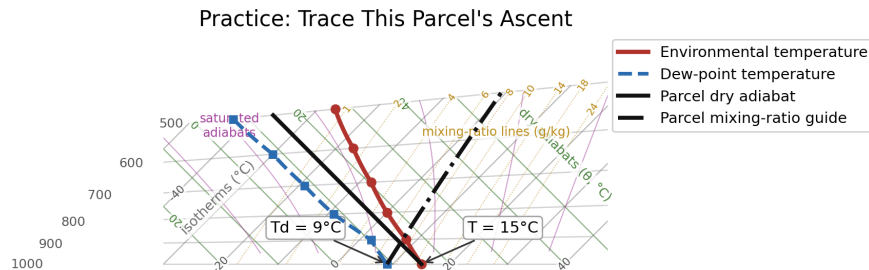
Step 1 – find the LCL. The two guide lines cross at about 915 hPa, where the parcel's dry-adiabatic temperature ($\approx 12.6^\circ\text{C}$) exactly equals the dew-point value implied by its fixed mixing ratio. Below this pressure the parcel's dew point is below its temperature (unsaturated); above it, continuing along the dry adiabat would push the implied dew point above the temperature, which cannot happen – so this crossing is the Lifting Condensation Level (LCL), where the parcel first saturates and cloud base would form.

Step 2 – switch to a saturated adiabat. From the LCL, the parcel continues rising along a *saturated adiabat* (magenta) rather than a dry adiabat: condensation now releases latent heat into the parcel as it cools, so it cools more slowly than an unsaturated parcel would. Following this saturated adiabat up to 700 hPa gives a parcel temperature of about 1.3°C .

Step 3 – compare with the environment. The environmental temperature at 700 hPa here is about -9°C (red curve). The parcel ($\approx 1.3^\circ\text{C}$) is therefore several degrees warmer than its surroundings, so it is positively buoyant at 700 hPa and would continue to accelerate upward under its own buoyancy (ignoring virtual-temperature and entrainment effects) – the same reasoning used to identify genuinely unstable layers on a real sounding.

YOU SHOULD TRY Question 8: Practice: trace a parcel's ascent on a tephigram

The figure below shows an environmental sounding, together with a surface parcel of $T = 15^\circ\text{C}$ and $T_d = 9^\circ\text{C}$ at 1000 hPa. Its parcel dry adiabat (solid black) and mixing-ratio guide (dash-dot black) are drawn from the surface, exactly as in the worked example above.



1. By eye, estimate the pressure at which this parcel reaches its LCL.
2. From the LCL, the parcel continues along a saturated adiabat. Estimate its temperature at 700 hPa.
3. The environmental temperature at 700 hPa is -8°C . Is the parcel positively or negatively buoyant there?
4. Explain, in one or two sentences, why the parcel's path follows a different family of guide lines above the LCL than below it.

Worked answer

The dry adiabat and mixing-ratio guide cross at approximately 910–915 hPa, which is the LCL (allow a reasonable graphical range, e.g. 895–930 hPa).

Following a saturated adiabat from the LCL to 700 hPa gives a parcel temperature of roughly -4 to -5°C .

Since the environment at 700 hPa is -8°C , the parcel (around -4 to -5°C) is warmer than its surroundings and is therefore positively buoyant.

Below the LCL the parcel is unsaturated, so it cools at the dry adiabatic lapse rate (following a dry adiabat) while its mixing ratio r_v stays fixed, so its dew point falls along a mixing-ratio guide. Once temperature and dew point meet at the LCL the parcel is saturated: any further ascent causes continuous condensation, which releases latent heat and slows the cooling rate, so the parcel instead follows a saturated adiabat.